

CISC 3667: Game Design and Development

In-Class Activity 2: Build a Game

Assigned: Thursday, September 10

Objective

In this activity, you will design a simple game using a limited set of **game elements, mechanics, and rules**. Your goal is to think about how the different parts of a game work together to create an experience for the player.

You will identify:

- Goals
- Mechanics
- Rules
- Objects
- Interactions
- Attributes
- States

You do **not** need to create or program the game. Your task is to **design a simple game concept**.

Part 1: Form Your Design Team

Work in a group of **4-5 students**.

Your group will design a simple game using a limited set of design constraints.

Your game can be:

- A board game
- A card game
- A video game
- A physical game
- Another type of playable game

The goal is not to create a large or complicated game. Instead, focus on creating a game that is:

- Understandable
- Playable
- Built around clear mechanics
- Governed by clear rules

Part 2: Your Design Constraints

Your challenge is to create a game using a **limited set of mechanics and objects**.

Choose **2 mechanics** and **3-4 objects** from the lists below.

Possible Mechanics

- Move
- Collect
- Avoid
- Shoot
- Match
- Trade
- Build
- Hide
- Race
- Jump
- Choose
- Push

Possible Objects

- Player
- Enemy
- Coin
- Key
- Door
- Timer
- Platform
- Card
- Resource
- Obstacle

You may decide how these elements work together, but you should **not simply add unlimited new mechanics**.

The challenge is to create an interesting game using limited ingredients.

Part 3: Design Your Game

1. Game Title

Give your game a name.

Game Title:

2. Goal

What is the player trying to accomplish?

The goal of our game is:

3. Core Mechanics

What does the player repeatedly do while playing?

Identify the two mechanics your group selected and explain how they work in your game.

Core Mechanic 1

Mechanic: _____

How does it work?

Core Mechanic 2

Mechanic: _____

How does it work?

How do these mechanics work together during gameplay?

4. Rules

What are the most important rules that define how your game works?

Think about:

- What can the player do?
- What can't the player do?
- What happens when the player succeeds or fails?
- How does the game begin?
- How does the game end?

You do **not** need to list every possible rule. Focus on the rules a player needs to understand in order to play.

Rule	How does it affect gameplay?
1.	
2.	
3.	
4.	
5.	
6.	
7.	

Part 4: Identify Your Game Objects

List the important objects or entities in your game.

Object	Role in the Game

Examples might include:

- Player
- Enemy
- Collectible
- Platform
- Card
- Resource
- Obstacle

Part 5: Interactions

Games are made up of elements that interact with one another.

Identify at least **two important interactions** in your game.

Think about:

Object/Player A → interacts with → Object B

For example:

Interaction	What Happens?
Player → Coin	The player's score increases.
Player → Enemy	The player loses health.
Player → Key	The player can unlock a door.

Your Game

Interaction	What Happens?

Part 6: Attributes

Choose at least **two objects** and identify important attributes associated with them.

An **attribute** is a characteristic or piece of information associated with an object.

For example:

Object	Possible Attributes
Player	Health, speed, score
Enemy	Health, movement speed, damage
Coin	Value
Door	Locked/unlocked

Your Game

Object	Attribute(s)

Part 7: States

Identify at least **two states** that can change during your game. For example:

- Playing / Paused
- Alive / Defeated
- Locked / Unlocked
- Hidden / Visible
- Day / Night
- Level 1 / Level 2

Your Game

Object/Game Element	Possible States

Part 8: Draw Your Game

Create a quick sketch, state diagram, or wireframe showing how your game works. Your drawing does **not** need to be artistic.

You might show:

- The player
- Important objects
- The game space
- How objects interact
- A basic game screen
- The flow of the game

The goal is to help someone else understand your game.

Part 9: The 30-Second Test

Imagine someone starts playing your game. Describe what happens during the **first 30 seconds of gameplay**.

Consider:

- What does the player see?
- What does the player do?
- What decisions do they make?
- What do they interact with?
- What are they trying to accomplish?

A player begins by...

Part 10: Share Your Game

Be prepared to briefly share your game with the class.

Your group should explain:

1. **What is your game?**
2. **What is the player's goal?**
3. **What does the player repeatedly do?**
4. **What are the most important rules?**
5. **What objects are important to the game?**
6. **How do those objects interact?**
7. **What attributes or states can change during gameplay?**

Design Challenge

Before you finish, discuss the following question with your group:

If we removed one of our core mechanics, would the game still work?

If the answer is **yes**, discuss how removing that mechanic would change the player's experience.

If the answer is **no**, explain why that mechanic is essential to your game.